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COVER-AFTER PILE GRAIN COVER

Abstract

A cover-after grain tarp is disclosed. The cover-after grain tarp includes, for example, a tarp body having a plurality of tarp sections joined together at one or more seams, each tarp section comprising a plurality of tarp panels having an outer terminal edge spaced between opposing converging edges and an edge panel secured along its length to an adjoining one of the plurality of tarp panels. The edge panel is configured for securing to hardware of a center tower structure at a grain storage location.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] The present disclosure relates to a cover for temporarily storing agricultural grains. More

particularly, but not exclusively, the present disclosure relates to a cover-after grain cover or yokeless cover-after grain cover, such as a tarp, for covering piles of agricultural grains.

BACKGROUND

[0002] Coverings, such as cover-after grain tarps, are used to cover grain (e.g., corn, soybean, milo, and wheat) that is temporarily stored at a grain storage site or other suitable locations. In the case of a cover-after tarp, the tarp is typically installed at the grain storage site over the grain pile. The cover-after grain cover generally takes the shape of the pile it covers. For example, in the case of round piles, cover-after grain covers are generally conical in shape and used to cover a generally conically shaped pile of grain. Grain cover tarps typically have seams joining several sections of material and transitional seams joining materials can be different by type, thickness, and quality. Collectively these materials form an assembled grain covering. Stresses carried by the grain cover can cause failure at seams, in the body of a single material, and between different components and/or assembled materials. For example, the yoke portion of a tarp is often difficult to manufacture and more likely to fail itself and/or at its one or more seams. Moreover, a cover's design, material components, and material strengths and cost can be a significant driver in the weight, construction, assembly, and installation of cover-after grain covers. Material and design changes in cover-after round piles can drastically affect installation, usability, and durability of the cover. Options for securing a cover can vary based on the type and availability of materials, design, and construction of the cover and installation location. Installing a cover as well as properly and adequately securing the cover to available structure at the installation location can be challenging. Therefore, what is needed is a cover-after grain cover that addresses these and other deficiencies in the cover-after grain cover industry.

SUMMARY

[0003] Therefore, it is a primary object, feature, or advantage of the present disclosure to improve over the state of the art.

[0004] It is a further object, feature, or advantage of the present disclosure to provide a grain cover that is configured to address and remedy material and seam failures in existing grain covers.

[0005] It is a still further object, feature, or advantage of the present disclosure to provide a cover-after grain cover that increases manufacturability and decreases material costs.

[0006] Another object, feature, or advantage is to provide a cover-after grain cover that reduces the chances of failure by better distributing stresses carried by the material and seams.

[0007] Yet another object, feature, or advantage is to provide a cover-after grain cover that decreases the time, labor, and potential for problems during installation.

[0008] Still another object, feature, or advantage is to provide a cover-after grain cover that includes features for easily securing the cover to structures at the grain pile, such as one or more structural members of a center tower.

[0009] It is yet a further object, feature, or advantage to provide a cover-after grain cover that is manufactured and assembled for increasing the ease and success of installation.

[0010] It is still a further object, feature, or advantage to provide a cover-after grain cover that is a yokeless cover-after grain cover.

[0011] It is another object, feature, or advantage to provide a cover-after grain cover that reduces the number of failure points and cost by being a yokeless cover-after grain cover.

[0012] At least one other object, feature, or advantage is to provide a cover-after grain cover that uses edge panels having different material characteristics to better carry and distribute stresses at seams and connection points for increasing the durability and longevity of the cover.

[0013] According to at least one aspect, a cover-after grain tarp is disclosed. The cover-after grain tarp includes, for example, a tarp body having a plurality of tarp sections joined together at one or more transition seams, each tarp section includes a plurality of tarp panels having an outer terminal edge spaced between opposing converging edges and an edge panel of each of the plurality of tarp sections is secured along a transition seam to a converging edge of an adjoining one of the plurality

of tarp panels in each of the plurality of tarp sections. The edge panel is configured for securing to one or more structural members of a tower structure at a grain storage location.

[0014] According to at least one other aspect, a yokeless cover-after grain tarp is disclosed. The yokeless cover-after grain tarp includes, for example, a tarp body having a plurality of tarp sections joined together at one or more transition seams, each tarp section includes a plurality of tarp panels having an outer terminal edge spaced between opposing converging edges and an edge panel of each of the plurality of tarp sections secured along a transition seam to an adjoining one of the plurality of tarp panels and the transition seam extending generally between the outer terminal edge and opposing converging edge of the adjoining one of the plurality of tarp panels. The edge panel is configured for securing to one or more structural members of a tower structure at a grain storage location.

[0015] According to another aspect, a method for a yokeless cover-after grain tarp is disclosed. The method for a yokeless cover-after grain tarp includes, such steps as, for example, providing a tarp body having a plurality of tarp sections with a plurality of tarp panels having an outer terminal edge spaced between opposing converging edges and each of the plurality of tarp sections having a transition seam connecting an edge panel to an adjoining one of the plurality of tarp panels and securing the edge panel to one or more structural members of a center tower structure at a grain storage location for securing the tarp body.

[0016] One or more of these and/or other objects, features, or advantages of the present disclosure will become apparent from the specification and claims that follow. No single aspect need provide each and every object, feature, or advantage. Different aspects may have different objects, features, or advantages. Therefore, the present disclosure is not to be limited to or by any objects, features, or advantages stated herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] Illustrated aspects of the disclosure are described in detail below with reference to the attached drawing figures, which are incorporated by reference herein, and where:

[0018] FIG. 1A is a pictorial representation showing a cover-after grain tarp in accordance with an exemplary aspect of the present disclosure;

[0019] FIG. 1B is a pictorial representation of FIG. 1A without a cover-after grain tarp for illustrating structural tower members supporting structural conveyor members of a grain conveyor within a grain bunker;

[0020] FIG. 2 is a pictorial representation of a section of a cover-after grain tarp;

[0021] FIG. 3A is a pictorial representation of top side of an edge panel and securement straps of a cover-after grain tarp;

[0022] FIG. 3B is a pictorial representation of an underside of an edge panel and

[0023] securement straps of a cover-after grain tarp;

[0024] FIG. 4 is a pictorial representation showing a top view of a cover-after grain tarp and installation atop a grain pile with a center tower structure in accordance with an exemplary aspect of the present disclosure;

[0025] FIG. 5 is another pictorial representation showing a top view of a cover-after grain tarp and installation atop a grain pile with a center tower structure in accordance with an exemplary aspect of the present disclosure;

[0026] FIG. 6 is another pictorial representation showing a top view of a cover-after grain tarp and installation atop a grain pile with a center tower structure in accordance with an exemplary aspect of the present disclosure;

[0027] FIG. 7 is another pictorial representation showing a top view of a cover-after grain tarp in

an unfurled position and installation of the cover-after grain tarp atop a grain pile with a center tower structure in accordance with an exemplary aspect of the present disclosure;

[0028] FIG. **8** is a pictorial representation showing a top view of joined sections of an installed cover-after grain tarp in accordance with an exemplary aspect of the present disclosure;

[0029] FIG. **9** is an enlarged view taken along line **9-9** in FIG. **8** of edge panel portions of a cover-after grain tarp and securement straps secured to structural members of a center tower structure in accordance with an exemplary aspect of the present disclosure;

[0030] FIG. **10** is a pictorial representation of an edge panel of a cover-after grain tarp and a securement strap partially tied to a structural member of a center tower structure in accordance with an exemplary aspect of the present disclosure;

[0031] FIG. **11** is another pictorial representation of an edge panel of a cover-after grain tarp and securement strap tied to a structural member of a center tower structure in accordance with an exemplary aspect of the present disclosure;

[0032] FIG. **12** is a pictorial representation of a flow diagram illustrating a method for assembling and installing a cover-after grain tarp; and

[0033] FIG. **13** is a pictorial representation of another flow diagram illustrating a method for assembling and installing a yokeless cover-after grain tarp.

DETAILED DESCRIPTION

[0034] FIGS. **1-13** illustrate exemplary aspects of a cover-after grain tarp **100** and methods of installation of the same also referred herein as a cover-after grain cover and a yokeless cover-after grain tarp/cover and their installation methods used for covering piles of stored grain (e.g., corn, soybean, milo, and wheat). Types of cover-after grain tarp **100** materials and thicknesses can vary. Material selection and thickness can depend on a range of factors, such as, for example, the use environment, application, desired life and durability, cost, and the material location (e.g., edge panel, tarp body, tarp panels, etc.) within the cover-after grain tarp **100**.

[0035] Cover-after grain tarp **100** is generally used in combination with a barrier wall **102** for forming a bunker **104** for housing a grain pile **105**. An outer terminal edge **108** of the body **106** of the tarp **100** may be secured to the barrier wall **102** for covering grain pile **105** underneath and securing cover-after grain tarp **100** atop grain pile **105**. In at least one aspect, outer terminal edge **108** is approximated by the collective outer edges of each tarp panel **142**. Barrier wall **102** is typically constructed from concrete or steel for forming bunker **104**. Battening (not shown) is generally attached to the upper portion of barrier wall **102** for securing outer terminal edge **108** of cover-after grain tarp **100** to the barrier wall **102**.

[0036] Grain is introduced into bunker **104** to create grain pile **105** surrounded and supported in a generally conical, vertically rising shape by barrier wall **102**. In many instances, grain is introduced into bunker **104** using a grain conveyor **110** supported by one or more (generally horizontal) structural conveyor members **112** and one or more (generally vertical) structural tower members **114**. Grain travels on grain conveyor **110** supported by structural conveyor members **112** and descends vertically through structural tower members **118** thereby forming conical grain pile **105** within bunker **104**. Cover-after grain tarp **100** is installed after bunker **104** is full and grain pile **105** is completely formed.

[0037] Center tower skirt **116**, which grain is fed through into conically shaped grain pile **105** below, may be secured around structural tower members **114**. Center tower skirt **116** is generally shaped to a configuration or collective shape of the structural tower members **114**. Center tower skirt **116** has an outer circumference extending between an upper edge **118** and a lower edge **120**. The upper edge **118** of the center tower skirt **116** is operably secured to the structural tower members **114** and/or structural conveyor members **112**. The lower edge **120** of the center tower skirt **116** is operably secured to the structural tower members **118**.

[0038] Grain pile **105** within bunker **104** is generally conical in shape and generally has a preferred angle, or an angle of repose for the specific type of grain material, of approximately 21 degrees or

thereabout relative to a horizontal plane, when the bunker **104** is full. The angle can vary, such as, for example, less than 21 degrees (e.g., 18, 19, 20 degrees) or more than 21 degrees (e.g., 22, 23, 24 degrees).

[0039] The center tower skirt **116** can be an optional feature based on the type of grain conveyor **110** system at bunker **104**. For example, a center tower formed from structural tower members **114** may include mechanical steel doors instead of center tower skirt **116** for distributing grain at different elevations along the vertical length of structural tower members **114**. As best illustrated in FIG. **1B**, structural tower members **114** may be configured as a center tower structure with a top end **124** operably secured to structural conveyor members **111** and a bottom end **126** operably supported by a structural support pad **128** configured atop ground surface **130**. In this configuration, grain travels along conveyor **110** supported by structural conveyor members **112** and is fed through structural tower members **114**, from the top end **124** toward the bottom end **126** forming conically shaped grain pile **105** expanding horizontally outward toward barrier wall **102** and vertically upward toward structural conveyor members **112**.

[0040] The body **106** of tarp **100** may be assembled from multiple tarp sections **132** of a singular type of tarp material, which are typically separately manufactured and joined at a tarp section seam **140** during an assembly process to form the cover-after grain tarp **100**. Different types of tarp material may be used for multiple tarp sections **132** and joined at a tarp section transition seam **140**. Although quarter (four) tarp sections **132** are shown, the present disclosure contemplates multiple tarp sections **132** forming body **106** of tarp **100**, such as a number of tarp sections **132** more or less than the number of tarp sections **132** shown. In one aspect, body **106** of tarp **100** may include four or less tarp sections **132**. In another aspect, body **106** may include 4 or more tarp sections **132**. Each tarp section **132** has a section outer edge **134** spaced between section converging edges **136**, **138**. The section outer edge **134** forms a portion, such as a quarter portion, of outer terminal edge **108** of body **106** of tarp **100**. Section outer edge **134** for each tarp section **132** is generally formed by outer panel terminal edges **156** and portions of outer panel edges **152**, wherein the lengths of the portions of outer panel edges **152** increase with the increasing distance of each tarp panel **142** from section converging edge **136**. Section converging edge **136** is formed by inner panel edge **154** of edge panel **144**. Section converging edge **138** of each tarp section **132** is formed by the combined inner panel terminal edge **158** of edge panel **144**, inner panel terminal edges **158** of adjoining tarp panels **142**, and inner panel terminal edge **158** of end panel **145**.

[0041] Grain cover tarps may include a yoke (not shown) with a center opening surrounding and connected to center tower skirt **116**, while outer edges of the yoke are connected to each tarp section. Cover-after grain tarp **100** is yokeless, meaning the body of the tarp includes no yoke or yoke materials, there is no yoke or yoke materials surrounding and connected to center tower skirt **116**, and there is no yoke or yoke materials connected to quarter tarp sections **132** of tarp **100**. The body **106** of the tarp **100** is further detailed in the proceeding description.

[0042] For purposes of the disclosure, a seam is defined as the joint between and joining of two of the same types of material, whereas a transition seam is defined as the joint between and the joining of two different types of material. Joints between two types of the same or different materials can be joined or connected using stitching, a heat weld (i.e., welding together with heat), and/or securing together with an adhesive, securement straps, or other means appropriate for types of material being joined together.

[0043] For example, FIG. **2** provides an illustration presenting a tarp quarter section **132** of the body **106** of a cover-after grain tarp **100** for simplification purposes. In a preferred aspect, the other sections of body **106** of tarp **100** are identical (e.g., see, for example, FIGS. **7-8**) and thus are not illustrated in FIG. **2**. In another aspect, the other sections of the body **106** of the tarp **100** can be configured differently from one another, if and as needed. Tarp quarter section **132** of tarp **100**, as shown in FIG. **2**, is constructed from a series of individual tarp panels **142** (see also FIG. **12**, Step **180**). In one aspect, one of the series of individual tarp panels **142** can be an edge panel **144**. In one

aspect, one of the series of individual tarp panels **142** can be an end panel **145**. In one aspect, tarp section **132** can be assembled with the outermost tarp panel **142** of each tarp section **132** being edge panel **144** and the innermost tarp panel **142** of each tarp section **132** being end panel **145** (see also FIG. 12, Step **180**). Each tarp panel **142**, each edge panel **144**, and each end panel **145** includes an outer panel edge **152** and an inner panel edge **154** spaced between an outer panel terminal edge **156** and an inner panel terminal edge **158**. Each tarp panel **142**, each edge panel **144**, and each end panel **145** have a panel length defined between their opposing outer panel terminal edge **156** and inner panel terminal edge **158**. Each tarp panel **142**, each edge panel **144**, and each end panel **145** have a panel width defined between their opposing outer panel edge **152** and inner panel edge **154**. The width of each tarp panel **142**, each edge panel **144**, and each end panel **145** can be the same. In another aspect, the width of one or more tarp panels **142**, edge panel **144**, or end panel **145** in a tarp section **132** can be different. Each tarp panel **142** (inclusive of end panel **145**) decreases in length with its increasing distance from section converging edge **136** thereby converging section outer edge **134** and section converging edge **138**. Outer edge **134** of each tarp section **132** approximates a portion of the circumference of barrier wall **102**. In at least one aspect, the convergence of section converging edges **136** and **138** is less than a 90-degree portion of the circumference of a tarp quarter section **132** for best approximating the conical shape of grain pile **105** while aligning section outer edge with barrier wall **102**. In other words, the angle between section converging edges **136** and **138** is less than 90-degrees for best approximating the conical shape of grain pile **105** while aligning section outer edge with barrier wall **102**. Snugly fitting each tarp quarter section **132** and thereby body **106** of cover-after grain tarp **100** to the topography of grain pile **105** results in the least amount of excess or slack tarp material, which decreases costs and the potential for tarp material failure or seam failure as a result of a maldistribution of stresses across a tarp panel transition seam **146**, tarp panel seam **148**, tarp panel **142**, edge panel **144**, end panel **145**, tarp section **142**, and/or body **106** of cover-after grain tarp **100**. Although a tarp section having an angle less than 90 degrees between section converging edges **136** and **138** is shown, the present disclosure contemplates angles greater than 90 degrees, if and as needed, to accommodate grain piles of varying shapes, such as a grain pile having a grain pile angle less than the angle of repose.

[0044] Adjoining tarp panels **142** (inclusive of edge panel **144** and end panel **145**) in each tarp section **132** are connected along their respective outer panel edge **152** and inner panel edge **154** generally between outer panel terminal edge **156** and inner panel terminal edge **158**. In one aspect, each successive adjoining tarp panel **142** (inclusive of edge panel **144** and end panel **145**) in each tarp section **132** are generally shorter in length. As a result, outer portions of outer panel edges **152** of each tarp panel **142** (inclusive of edge panel **144** and end panel **145**) generally adjacent section outer edge **134** are unconnected to outer portions of inner panel edges **154** of each tarp panel **142** (inclusive of edge panel **144** and end panel **145**) generally adjacent section outer edge **134**. The length of unconnected portions can be configured to increase with the increasing distance of each tarp panel **142** from section converging edge **136**. Adjoining tarp panels **142** (inclusive of edge panel **144** and end panel **145**) in each tarp section **132** are connected along their respective inner panel edge **154** and outer panel edge **152** generally between outer panel terminal edge **156** and inner panel terminal edge **158**. Adjoining tarp panels **142** (inclusive of end panel **145**) are connected by a tarp panel seam **148**. An adjoining tarp panel **142** of one type of tarp material and edge panel **144** of another type of tarp material are connected by a tarp panel transition seam **146**. In another aspect, an adjoining tarp panel **142** of one type of tarp material and edge panel **144** and end panel **145** of same type of tarp material are connected by a tarp panel seam **146**. In at least one aspect, edge panel **144** is longer than tarp panels **142** (inclusive of end panel **145**) in each tarp section **132**. In at least one aspect, end panel **145** is shorter than tarp panels **142** and edge panel **144** in each tarp section **132**. Outer panel edge **152** of end panel **145** can be configured as a terminal outer edge forming a portion of section outer edge **134** for each tarp section **132**. Additionally, as shown in FIG. 2, end panel **145** is another one of tarp panels **142** and of the same material. In at

least one other aspect, end panel **145**, like edge panel **144**, can be of a different type of tarp material than tarp panels and connected to the adjoining tarp panel **142** by a tarp panel transition seam **146**. [0045] As best shown in FIG. 3A and FIG. 3B, edge panel **144** includes securement straps **160**. Each securement strap **160** has a secured end **162** and a free end **164**. Secured end **162** of each securement strap **160** is attached to edge panel **144** (see also FIG. 12, Step **182**). When cover-after grain tarp **100** is installed, free end **164** of each securement strap **160** is tied to one or more structural tower members **114**, as best illustrated in FIGS. 9-11. In at least one aspect, secured end **162** is sewn to tarp material, such as with a stitching **168** material, illustrated, for example, in FIGS. 10-11 (see also FIG. 12, Step **182**). At least one securement strap **160** has its secured end **162** attached to edge panel **144** generally parallel to and proximate inner panel edge **154** with its free end **164** extending generally outward from and perpendicular to inner panel terminal edge **158** of edge panel **144**. At least one securement strap **160** has its secured end **162** attached to edge panel **144** generally parallel to and proximate inner panel terminal edge **158** with its free end **164** extending generally outward from and perpendicular to inner panel edge **154** of edge panel **144**. The at least one securement straps **160** can each be configured having an angle relative inner panel edge **154** and inner panel terminal edge **158** of less than 90 degrees and greater than 45 degrees. A pair of securement straps **160** have their secured ends **162** attached to edge panel **144** generally angled to and proximate inner panel edge **154** with their free ends **164** extending generally outward from and angled to inner panel edge **154** of edge panel **144**. The pair of securement straps **160** can have an angle relative to inner panel edge **154** of less than 90 degrees and greater than 45 degrees. The pair of securement straps **160** can also configured perpendicular to inner panel edge **154** and parallel to inner panel terminal edge **158**. In at least one aspect, free ends **164** of securement straps **160** extend outward from an underside **170** of edge panel **144** to prevent malalignment of stresses on edge panel **144**, inner panel edge **154**, inner panel terminal edge **158**, and adjoining tarp panels **142**. The present disclosure contemplates that free ends **164** of securement straps **160** can extend from the topside **172** of edge panel **144**, if and as needed. Supporting straps **166** for aiding in the attachment of secured ends **162** of securement straps **160** to edge panel are illustrated in FIG. 3B, which shows the underside **170** of edge panel **144** for each tarp section **132**. Secured ends **162** of securement straps can be attached to edge panel **144** by sandwiching edge panel **144** between secured ends **162** and supporting straps **166** as best illustrated in FIG. 3A and FIG. 3B. Stitching **168** (see FIG. 3B) can be sewn through secured ends **162** and edge panel **144** for attaching securement straps **160** to edge panel **144**. The present invention contemplates securement straps attached to tarp panels **142**, if and as needed, such as in the case, if and when, there is no edge panel **144** employed in one or more tarp sections **132**.

[0046] As best shown in FIGS. 4-8, tarp sections **132** are assembled on top of grain pile **105** to form body **106** of cover-after grain tarp **100**. Prior to assembly atop grain pile **105**, tarp sections **132** can be accordion folded along tarp panel seams **148** with end panel **145** being located on top of the accordion fold and edge panel **144** being located on the bottom of the accordion fold (see also FIG. 12, Step **184**). Topside **172** of edge panel **144** faces upward when edge panel **144** is located on the bottom of the accordion folded tarp section **132**. Edge panel **144** is pulled out from its position at the bottom of the accordion folded tarp section **132**, folded over on top of the end panel **145** (underside **170** of edge panel **144** faces upward), and rested atop of end panel **145** at the top of the accordion folded tarp section **132** (see also FIG. 4 & FIG. 12, Step **186**) to facilitate and ease the transportation of each tarp section **132** atop grain pile **105**, securement of edge panel **144** to structural tower members **114** and unfolding of accordion folded tarp panels atop grain pile **105**. Securement straps **160** can be used to transport tarp sections **132** onto grain pile **105** and tug tarp sections up grain pile **105** to position securement straps **160** and edge panel **144** adjacent structural tower members **114** as illustrated, for example, in FIG. 4 (see also FIG. 12, Step **188**). With tarp sections **132** positioned atop grain pile **105**, edge panels **144** are unfolded from atop end panel **145** and into position closely adjacent structural tower members **114** as illustrated, for example, in FIG.

5 (see also FIG. 12, Step 190). Securement straps are positioned generally adjacent structural tower members 114 when tarp sections 132 are positioned atop grain pile 105 (see also FIG. 12, Step 192). Securement straps 160 are tied to structural tower members 114 as illustrated, for example, in FIGS. 9-11. In one aspect, each edge panel 144 includes at least one securement strap 160 tied to one of structural tower members 114 and at least one other securement strap 160 tied to a separate one of structural tower members 114 as illustrated, for example, FIG. 9. In one aspect, each edge panel 144 includes at least two securement straps 160 tied to one of structural members 114 and at least another two securement straps 160 tied to a separate one of structural tower members 114 as illustrated, for example, FIG. 9. In one aspect, each edge panel 144 includes at least two securement straps 160 with one of the two straps tied to one of structural tower members 114 and the other strap tied to securement strap 160 of an adjoining edge panel 144 and another two securement straps 160 with one of the another two straps tied to one of structural tower members 114 and the other strap tied to another securement strap 160 of an adjoining edge panel 144. Securement straps are tied to structural tower members 114 as shown, for example, in FIG. 10 and FIG. 11, by creating knot 169 and tying off a loop 167 in a portion of the strap of the free end, passing the end portion of the free end 164 of the strap around structural tower member 114, threading free end 164 of the strap back through the loop 167, and tying off free end 164 to a portion of the strap before the loop 167.

[0047] Tarp sections 132 positioned atop grain pile 105 and having edge panels 144 secured to structural tower members 114 as illustrated, for example, in FIG. 5, are unfurled from their accordion folded position by pulling end panels 145 toward barrier wall 102 (see also FIG. 12, Step 194). End panels 145 of each tarp section 132 are pulled until each tarp section 132 is completely unfurled whereby end panels 145 and section outer edges 134 of each tarp section are positioned adjacent or even partially overhang barrier wall 102 as illustrated, for example, in FIG. 7 (see also FIG. 12, Step 196). Section converging edges 138 of each tarp section 132 are also position adjacent edge panels 144 of adjoining tarp sections 132 as illustrated, for example, in FIG. 8 (see also FIG. 13, Step 198). Section outer edges 134 of each tarp section 132 are aligned with barrier wall 102 (see also FIG. 13, Step 200).

[0048] As best shown in FIG. 1A and FIG. 8, tarp sections 132 are connected to form body 106 of cover-after grain tarp 100. In one aspect, edge panels 144 of tarp sections 132 are connected to converging edge sections 138 of adjoining tarp sections 132 to form body 106 of cover-after grain tarp 100. Section converging edge 136 of one tarp section 132 is connected to section converging edge 138 of another tarp section 132 by a tarp panel transition seam 146 in the case where edge panels 144 are a different material from tarp panels 142 in each tarp section 132 (see also FIG. 13, Step 202). The present disclosure contemplates that section converging edge 136 of one tarp section 132 is connected to section converging edge 138 of another tarp section 132 by a tarp panel seam 148 in the case where edge panels 144 are the same material as the tarp panels 142 in each tarp section 132. In one aspect, tarp panel transition seam 146 is: (1) disposed in tarp sections 132 generally along and proximate to or at inner panel terminal edges 158 of edge panel 144, inner panel terminal edges 158 of adjoining tarp panels 142, and inner panel terminal edges 158 of adjoining end panel 145; and (2) disposed in adjoining tarp sections 132 generally along and proximate to or at outer panel edge 152 of edge panel 144. In at least one configuration of body 106 of cover-after grain cover 100, tarp panel transition seam 146 traverses: (1) generally between outer panel terminal edge 156 and inner panel terminal edge 158 of edge panel 144; and (2) generally between inner panel edge 154 of edge panel 144 and outer panel edge 152 of end panel 145. Outer terminal edge 108 of body 106 of cover-after grain tarp 100 can be secured to barrier wall 102 with battening (not shown) (see also FIG. 13, Step 204).

[0049] In a traditional grain cover that includes a yoke, only a few tarp panels of each tarp section are connected to the yoke and the yoke is connected to the structural tower members. Thus, the load of the tarp body is carried/transferred by the few panels of each tarp section and the seam

between the few panels connected to the yoke and to the structural tower members. Yoke material is often a different material, more expensive, more difficult to manufacture and assemble than panel material to adequately support the load shared between a few panels of each tarp section and the yoke. Nevertheless, failures often occur at the seam or material adjacent the seam between the few panels of each tarp section and the yoke. As best shown in FIG. 8, edge panel **144** is connected along tarp panel transition seam **146** to each panel **142** (inclusive of edge panel **144** and end panel **145**) of the adjoining tarp section **132**. Thus, the load of cover-after grain tarp **100** connected to structural tower members **114** is distributed evenly across all panels **142** (inclusive of edge panel **144** and end panel **145**) of the adjoining tarp section **132**. No single tarp panel **142** or a fewer number of tarp panels **142** than those making up a tarp section **132** carry the load, thereby evenly distributing the load of the tarp **100** from structural tower members **114** across the entirety of body **106**.

[0050] The disclosure is not to be limited to the particular aspects described herein. In particular, the disclosure contemplates numerous variations in a cover-after grain tarp or yokeless grain tarp **100**. The foregoing description has been presented for purposes of illustration and description. It is not intended to be an exhaustive list or limit any of the disclosure to the precise forms disclosed. It is contemplated that other alternatives or exemplary aspects are considered included in the disclosure. The description is merely examples of aspects, processes, or methods of the disclosure. It is understood that any other modifications, substitutions, and/or additions can be made, which are within the intended spirit and scope of the disclosure.

Claims

1. A cover-after grain tarp, comprising: a tarp body having a plurality of tarp sections joined together at one or more transition seams, each tarp section comprising a plurality of tarp panels having an outer terminal edge spaced between opposing converging edges; an edge panel of each of the plurality of tarp sections is secured along a transition seam to a converging edge of an adjoining one of the plurality of tarp panels in each of the plurality of tarp sections; wherein the edge panel is configured for securing to one or more structural members of a tower structure at a grain storage location.
2. The cover-after grain tarp of claim 1, wherein the edge panel has a length exceeding the length of the plurality of tarp panels in a tarp section.
3. The cover-after grain tarp of claim 1, wherein the edge panel comprises a separate type of material from the plurality of tarp panels.
4. The cover-after grain tarp of claim 1, wherein an outer converging edge of the edge panel of each of the plurality of tarp sections is secured at a transition seam along one of the opposing converging edges of an adjoining one of the plurality of tarp sections.
5. The cover-after grain tarp of claim 1, wherein the edge panel includes one or more securement straps generally disposed proximate a converging inner terminal edge of each edge panel for securing each edge panel of each of the plurality of tarps sections to the one or more structural members of the tower structure at the grain storage location.
6. The cover-after grain tarp of claim 1, wherein one of the opposing converging edges comprises an outer converging edge of the edge panel.
7. The cover-after grain tarp of claim 1, wherein one of the opposing converging edges comprises converging inner terminal edges of the plurality of tarp panels.
8. The cover-after grain tarp of claim 1, wherein the plurality of tarp panels are parallel.
9. The cover-after grain tarp of claim 1, wherein an angle between the opposing converging edges of each of the plurality of tarp sections is less than 90 degrees.
10. A yokeless cover-after grain tarp, comprising: a tarp body having a plurality of tarp sections joined together at one or more transition seams, each tarp section comprising a plurality of tarp

panels having an outer terminal edge spaced between opposing converging edges; an edge panel of each of the plurality of tarp sections secured along a transition seam to an adjoining one of the plurality of tarp panels and the transition seam extending generally between the outer terminal edge and opposing converging edge of the adjoining one of the plurality of tarp panels; wherein the edge panel is configured for securing to one or more structural members of a tower structure at a grain storage location.

11. The yokeless cover-after grain tarp of claim 10, wherein an outer converging edge of the edge panel of each of the plurality of tarp sections is secured at a transition seam along one of the opposing converging edges of an adjoining one of the plurality of tarp sections.

12. The yokeless cover-after grain tarp of claim 10, wherein the edge panel comprises a continuous panel of a first tarp material and the adjoining one of the plurality of tarp panels comprises a continuous panel of a second tarp material.

13. The yokeless cover-after grain tarp of claim 10, wherein the edge panel includes one or more securement straps generally disposed at a converging inner terminal edge and outer converging edge of the edge panel for securing to the one or more structural members of the tower structure at the grain storage location.

14. The yokeless cover-after grain tarp of claim 10, wherein one of the opposing converging edges comprises an outer converging edge of the edge panel and another one of the opposing converging edges comprises converging inner terminal edges of the plurality of tarp panels.

15. The yokeless cover-after grain tarp of claim 10, wherein the plurality of tarp panels are parallel and an angle between the opposing converging edges is less than 90 degrees.

16. A method for a yokeless cover-after grain tarp, comprising: providing a tarp body having a plurality of tarp sections comprising a plurality of tarp panels having an outer terminal edge spaced between opposing converging edges and each of the plurality of tarp sections having a transition seam connecting an edge panel to an adjoining one of the plurality of tarp panels; securing the edge panel to one or more structural members of a center tower structure at a grain storage location for securing the tarp body.

17. The method of claim 16, further comprising: securing the edge panel of each of the plurality of tarp sections to one of the opposing converging edges of an adjoining one of the plurality of tarp sections for assembling the tarp body.

18. The method of claim 16, further comprising: pulling an outermost tarp panel of the plurality of tarp panels away from the edge panel for expanding the tarp section atop the grain pile.

19. The method of claim 16, wherein the securing step comprises attaching one or more securement straps along a converging inner terminal edge of the edge panel and an outer converging edge of the edge panel to the one or more structural members of the center tower structure.

20. The method of claim 16, further comprising: accordion folding the plurality of tarp panels of each of the plurality of tarp sections together before installing at the grain storage location; and unfolding the edge panel atop a grain pile for revealing an outermost tarp panel of the plurality of tarp panels for installing the tarp body atop the grain pile.
